
Wordcount using pig:

```
lines = LOAD '/user/hadoop/HDFS_File.txt' AS (line:chararray);
words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) as word;
grouped = GROUP words BY word;
wordcount = FOREACH grouped GENERATE group, COUNT(words);
DUMP wordcount;
```

Matrix multiplication using pig:

```
--This file multiplies mat1*mat2
--The output has the same format as the two input matrices: (row,col,value)

mat1 = LOAD 'matrix1.csv' USING PigStorage(',') AS (row,column,value:int);
mat2 = LOAD 'matrix2.csv' USING PigStorage(',') AS (row,column,value:int);

A = JOIN mat1 BY column FULL OUTER, mat2 BY row;
B = FOREACH A GENERATE mat1::row AS m1r, mat2::column AS m2c,
(mat1::value)*(mat2::value) AS value;
C = GROUP B BY (m1r, m2c);

multiplied = FOREACH C GENERATE group.$0 as row, group.$1 as column, SUM(B.value)
AS value;

STORE multiplied INTO 'product' USING PigStorage(',');
```

Wordcount using hive:

```
SELECT word, COUNT(*) FROM input LATERAL VIEW explode(split(text, ' ')) lTable as
word GROUP BY word;

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CREATE TABLE docs (line STRING);
LOAD DATA INPATH 'text' OVERWRITE INTO TABLE docs;
CREATE TABLE word_counts AS
SELECT word, count(1) AS count FROM
(SELECT explode(split(line, '\s')) AS word FROM docs) w
GROUP BY word
ORDER BY word;
```